Fiscal Year:	FY 2019	**Task Last Updated:**	FY 01/03/2019
PI Name:	Laughlin, Mitzi Ph.D.		
Project Title:	Assessment of the Time Courses of Leg Muscle Strength and Performance Losses During Prolonged Spaceflight using MARES		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mslaughl@Central.UH.EDU	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	713-743-0405
Organization Name:	University of Houston		
PI Address 1:	Health and Human Performance		
PI Address 2:	3855 Holman St., Garrison Rm. 105R		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77204-6015	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Crew Health NNJ10ZSA003N
Start Date:	02/23/2016	**End Date:**	08/24/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 8/24/2020 per HHC element/JSC and NSSC information (Ed., 12/10/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clarke, Mark SF Ph.D. (University of Houston) Layne, Charles S Ph.D. (University of Houston)		
Grant/Contract No.:	NNX16AF63G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The grant NNX16AF63G Assessment of the Time Courses of Leg Muscle Strength and Performance Losses During Prolonged Spaceflight using MARES began in February 2016 and is approaching the third-year anniversary. The study is designed to examine declines in muscular strength, function, and neuromotor control during long-duration spaceflight using the Muscle Atrophy Research and Exercise System (MARES) device in the Columbus module on the International Space Station (ISS). The knowledge gaps addressed by this study relate to the risks to human space travelers associated with in-flight changes in muscle function (PRD Risk: Risk of Impaired Performance Due to Reduced Muscle Mass, Strength, and Endurance), as identified by Integrated Research Plan (IRP) Gaps M24 (What is the time course of changes in muscle protein turnover, muscle mass and muscle function during long-duration spaceflight?) and M2 (What is the current status of in-flight/post-flight exercise performance capability?). Specifically, our study seeks to characterize adaptive in-flight changes in lower limb muscle strength and neuromotor control performance using the Muscle Atrophy Research and Exercise System (MARES).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed investigation seeks to systematically examine the relationships among muscle strength and neuro-motor control performance. By tracking these relationships over time during flight we will be able to identify any adaptive changes in integrated motor performance. If successful, this investigation should provide significant benefits to the Human Research Program by quantifying the aspects of the Program Requirements Document (PRD) Risk of Impaired Performance Due to Reduced Muscle Mass, Strength and Endurance embodied in IRP (Integrated Research Plan) Gap M24 (What is the time course of changes in muscle protein turnover, muscle mass and muscle function during long-duration spaceflight?) and M2 (What is the current status of in-flight/post-flight exercise performance capability?). Additionally, this investigation will provide valuable information to clinicians regarding muscle atrophy due to injury or disease on Earth.
Task Progress:	Spaceflight induces significant loss of skeletal muscle mass and function in response to mechanical unloading of the neuro-musculoskeletal system. Contractile protein degradation, sarcomere rearrangement, reduced cross-sectional area, and consequent decrements in force production are all hallmarks of this response at the myofiber level. It is also well accepted that spaceflight induced myofiber alterations may be modulated or even triggered by a disruption in the interconnected “neuromuscular networks,” which normally exist between myofiber and neural elements of the musculoskeletal system under terrestrial loading conditions. Activation of these neuromuscular networks for the antigravity muscles of the lower limbs and trunk is strongly influenced by load receptors in the soles of the feet and the muscles and joints of the lower extremities. Thus, off-loading of these receptors by the microgravity environment of spaceflight has been suggested as an important triggering mechanism for the muscle atrophy and fiber-type shifting associated with prolonged spaceflight, as well as the alterations in sensorimotor function that contribute to post-flight disruption of balance and locomotor control. As the means to assess strength during flight have not frequently been available, most of the data used to formulate these hypotheses have come from post-flight assessments (relative to pre-flight values) and or ground-based simulations of off-loading (e.g., human bedrest studies and rodent hind-limb suspension studies). Availability of the MARES system allows us now, for the first time, to gather data regarding the time course of changes in muscle strength and performance during long-term spaceflight missions. Additionally, the MARES will enable us to assess potential adaptations of proprioceptive function throughout the mission and relate these adaptations to changes in muscle strength and function. Our primary objective is to gather data that are predominantly functional in nature, focusing on standardized measures of strength and performance to characterize for the first time the time course of changes in these functions during prolonged spaceflight. Hypotheses: Our fundamental hypothesis is that the time courses of adaptive changes in strength and performance of the anti-gravity muscles of the lower limb will follow exponential trajectories, with rapid changes occurring early in the mission followed by slower changes later in a mission. Specific Aims: We propose to test this hypothesis by using the MARES device to quantify the time course of changes in muscle strength and performance of crewmember volunteers during six-month ISS missions. We will establish baseline strength and performance capabilities of each participating crewmember from two pre-flight test sessions, track changes from that baseline during three in-flight test sessions, and track recovery to baseline levels during five post-flight test sessions. At each session we aim to: 1. Measure isometric and isokinetic strength of the ankle plantar flexor and knee flexor and extensor muscle groups using standard techniques programmed into the MARES device, and 2. Measure dynamic motor control performance during a set of position-matching tasks performed with and without external, MARES-imposed joint torque loads, 3. Measure the balance and potential adaptation of flexor and extensor motor (EMG) activations during strength and motor control performance testing. Study Progress: Extensive progress has been made in the third year and data collection has been completed. Four crewmembers completed all phases of MARES testing. The data quality from ground testing and onboard ISS was excellent. Data processing and analysis is underway. Abstracts have been submitted to the 2019 American College of Sports Medicine and the NASA Human Research Program (HRP) meetings. These abstracts detail the knee strength and function data during the pre and in-flight data stages. The post-flight data will be analyzed along with more data collected pre and post-flight only by the European Space Agency. Manuscripts are also being planned and written along with our scientific partners. Presentations: Rittweger J, Tomilovskaya E, Albracht K, Bottinelli R, Capri M, Cerretelli P, Clarke MSF, Flück M, Franceschi C, Gelfi C, Koryak Y, Layne CS, Narici M, Reed JT, Shenkman B, Stäudle B, Laughlin MS, Kozlovskaya I. Early changes in plantar flexor muscle strength during spaceflight: Results from the in-flight Sarcolab3 protocol. Centro Nacional de Estudios Espaciales (National Centre for Space Studies), Toulouse, France, September 19, 2018. Laughlin MS, Reed JT, Layne C, Kozlovskaya I, Koryak Y, Albracht K, Staeudle B, Rittweger J. SARCOLAB-3: Myotendinous and Neuromuscular Adaptation to Long-term Spaceflight. NASA Human Research Program Investigators' Workshop – The Gateway to Mars, Galveston, Texas, January 22-25, 2018.

Bibliography Type:	Description: (Last Updated: 12/10/2020)
Abstracts for Journals and Proceedings	Rittweger J, Tomilovskaya E, Albracht K, Bottinelli R, Capri M, Cerretelli P, Clarke MSF, Flück M, Franceschi C, Gelfi C, Koryak Y, Layne CS, Narici M, Reed JT, Shenkman B, Stäudle B, Laughlin MS, Kozlovskaya I. "Early changes in plantar flexor muscle strength during spaceflight: Results from the in-flight Sarcolab3 protocol." Centro Nacional de Estudios Espaciales (National Centre for Space Studies), Toulouse, France, September 19, 2018. Centro Nacional de Estudios Espaciales (National Centre for Space Studies), Toulouse, France, September 19, 2018. , Sep-2018
Abstracts for Journals and Proceedings	Laughlin MS, Reed JT, Layne C, Kozlovskaya I, Koryak Y, Albracht K, Staeudle B, Rittweger J. "SARCOLAB-3: Myotendinous and Neuromuscular Adaptation to Long-term Spaceflight." 2018 NASA Human Research Program Investigators' Workshop, The Gateway to Mars, Galveston, TX, January 22-25, 2018. 2018 NASA Human Research Program Investigators' Workshop, The Gateway to Mars, Galveston, TX, January 22-25, 2018. , Jan-2018